Invoice Desk

An AI agent for accounts-payable invoice processing · Zamp AI Solutions Analyst case study, problem 1

Built by: Naveen Prakash Junodiya · BS student, IIT Madras · Roll no. BS23B011 · November 2026

What	Link
Live app (runnable)	invoice-desk-production-8d50.up.railway.app
Demo video (5 min)	loom.com/share/5420ebc4ea244d83ae0ef0c1197d3a06
Code, tests, docs	github.com/bs23b011-commits/invoice-desk

Please note: the email part (n8n + Gmail) works only on my laptop
In the video, the agent emails Stores, I reply from Gmail, and n8n hands the reply back to the agent. That email loop runs on my laptop, where n8n and my Gmail account are connected. **On the live link, no real emails are sent.** Everything else works the same online: every stage, every check and every decision. Each message the agent would send is shown in the invoice's Discussion panel and under Reports → Activity, and you can answer it with the **Try a reply (demo)** box.

What it does

Invoice Desk reads each invoice, checks it against itself, our records and Indian tax rules, asks the right person when something is missing, and recommends **approve, hold or reject** with its reasons. **It never pays on its own:** a person makes the final call on every payment.

What the brief asks for, and where it is

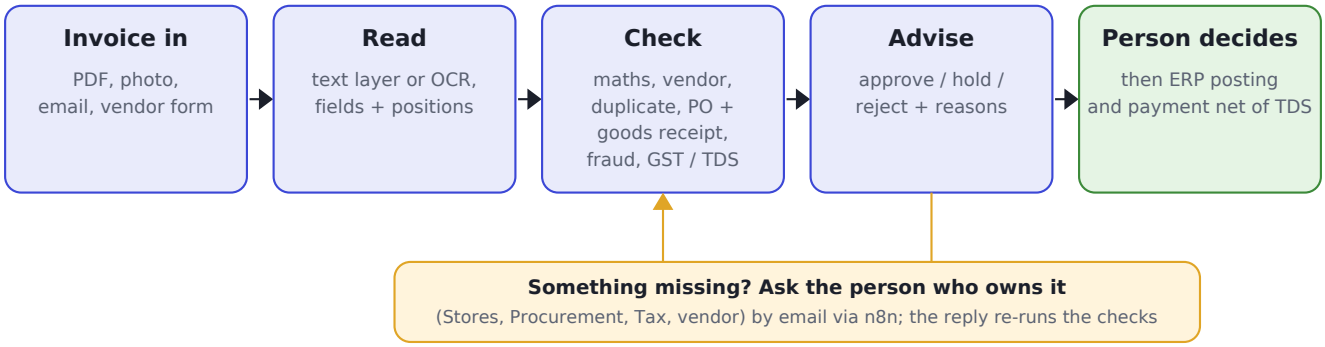
The brief asks for	Where it is in Invoice Desk
Accepts a real input	**Upload & check**: any PDF or phone photo, including scans and several invoices in one PDF. On my laptop also by email and a vendor upload form through n8n.
Runs through real logic	11 stages: read, extract, self-check, vendor, duplicate, PO and goods receipt, fraud, tax, recommendation, then ERP posting after a person approves.
Produces a real output	A recommendation with reasons, the questions sent to people, an ERP journal, and a payment scheduled net of TDS.
UI showing each stage	Each stage ticks live beside the invoice; clicking a field highlights where it was read on the page.
Dashboard of past runs	**Reports**: every run, what the agent advised, what the person decided, PO usage and accuracy.
Video: happy path + edge case	Clean invoice; goods not received (email loop); new bank account (fraud); credit note.

The problem, and why it is hard

Before paying an invoice, an AP clerk checks that it is real, correct, ordered, received and taxed right. Reading the PDF is the easy part. The hard part is everything around it:

- **Invoices don't look alike.** Every vendor has its own layout, plus phone photos, faxes, sideways scans and several invoices in one PDF.
- **The answer is spread across teams.** The PO is with Procurement, the goods receipt with Stores, the vendor's bank with the master data, the tax rules with Finance.
- **The expensive mistakes are quiet.** A credit note paid out, an advance paid twice, a re-sent duplicate, a fraudster's "new bank account". Each looks like a normal invoice.
- **Indian tax adds its own rules.** GSTIN checks, CGST/SGST vs IGST, e-invoice IRN, TDS, the MSME 45-day limit, and the GST 2.0 slab change of 22 Sep 2025.
- **Most problems need a person.** Someone has to confirm the goods arrived or the bank changed, and someone has to chase them.

How it works



The agent runs every check and explains each one. A missing fact becomes **one question to the person who owns it**, and their reply re-runs the checks. Payment only happens after a person approves; then it is posted to the ERP, read back to confirm, and paid net of TDS.

The checks, in plain words

Stage	What it checks
Read	Uses the PDF's text, or OCR for scans (turns sideways pages upright, ignores junk text layers and hidden white text).
Self-check	The invoice must add up by itself: quantity × rate per line, lines to subtotal, tax to total, and the amount in words. A misread digit breaks one of these, so the agent knows when it misread.
Vendor	GSTIN valid and on file, billed to our GSTIN, vendor not blocked, look-alike names caught.
Duplicate	Same invoice sent again, even in a different layout or with a reworded number.
3-way match	Price and quantity match the purchase order; Stores has received what is billed; the PO isn't used twice.
Fraud	Bank account matches the one on file; hidden instructions; e-invoice QR matches the printed total.
Tax	Right GST type and slab for the date, e-invoice number, TDS to deduct, MSME 45-day payment limit.

Edge cases it handles

Situation	What the agent does	Why
Goods billed before they arrived (300 billed, 200 received)	Holds, emails Stores, reads the reply, records the receipt and re-checks	Paying for goods not received is a direct loss
New bank account on a real vendor's invoice	Holds; never emails the vendor; approval needs "I called them on the number on file"	The vendor's inbox may be the hacked one
Look-alike vendor name with another GSTIN	Rejects as impersonation	Classic fake-vendor fraud
Duplicate, re-sent with another layout or a transposed number	Rejects, or holds when only amount and date match	Paying twice
Credit note, proforma, quotation, statement	"Not a bill to pay": no approve button, only dismiss	Tools that only read numbers have paid these out
Advance already paid, deducted after the total	Holds; if approved, pays only the balance	Otherwise the advance is paid twice
Amount in words disagrees with the figures	Holds	A typo or an altered figure
12% GST on an invoice dated after 22 Sep 2025	Holds: that slab no longer exists after GST 2.0	Vendor billing software not updated
Hidden white text telling an AI to approve	Ignores it, flags it, asks for a phone check	Prompt injection aimed at AI agents
Sideways scan; scan with a junk text layer	Turns it upright / reads the image instead	Real scanners do this every day
E-invoice QR doesn't match the printed total	Holds as altered after issue	The signed QR can't be edited, the page can
Two invoices against one PO before either is approved	Second one held for the PO balance	The PO would be used twice
Several invoices in one PDF, two vendors mixed	Splits them, counts copies once, holds mixed files	Batch scans are common
Invoice from a vendor not in our records	Holds as "vendor not onboarded"; testers can use Test setup	Never pay an unknown vendor

The full list of 32 hard cases, each with the real complaint it came from, is in the repo: docs/HARD_CASES.md.

How I built it

- **Started from real complaints.** I read public reviews and forum posts about AP tools (Ramp, Stampli, BILL and others), OCR failure write-ups and Indian GST/TDS audit findings, and listed what actually goes wrong in practice.
- **Turned each complaint into a test document.** That became the 32-case hard set (credit notes, advances, sideways scans, hidden text, GST 2.0) plus 50 invoices with known answers and 20 messy scans.
- **Built the agent so every decision can be tested.** AI-style reading handles messy paper; the pay / don't-pay decision is plain rules, the same every time, with automated tests for each one.
- **Built the UI around the person who decides.** Each card says what the agent advises and why, what happens if you approve, and who is being asked.
- **Connected the real world with n8n.** Email intake, a vendor upload form, notifications, an escalation timer and reply reading, tested end to end with my own Gmail on my laptop.
- **Deployed it** with Docker on Railway, so anyone can run it from a link.

Tech stack

Part	Built with
Backend and API	Python, FastAPI, SQLite
Reading invoices	pdfplumber and pypdfium2 for digital PDFs; Tesseract OCR with orientation detection for scans; zxing-cpp for e-invoice QR codes
Decisions	Plain Python rules: arithmetic identities, 3-way match, fraud and Indian tax checks (GST 2.0, TDS, MSME)
Interface	Plain HTML, CSS and JavaScript; Jeen, an assistant that answers from the data but cannot approve or pay
Messaging	n8n (5 workflows) + Gmail, on my laptop
Hosting	Docker on Railway (live link); code on GitHub

Proof

Test	Result
50 generated invoices with known answers (10 situations, 3 layouts, digital and fax-quality scans)	**100%** right decisions, **0** wrong approvals
125 single-digit misreads injected into scans	**100%** caught by the self-check
Hard set: 32 documents built from real complaints	**32 / 32** handled as a careful AP clerk would
Messy set: 20 bad scans and odd layouts	**20 / 20**
Jeen: 31 question types	**31 / 31**
Automated tests	**286** passing
Speed	about **0.6 s** per digital invoice

Careful on purpose: 5 of the 10 good fax-quality scans were sent to a person, because OCR was unsure of the invoice number. Holding is cheap; paying wrongly is not.

Design choices, and why

- **AI reads, rules decide.** Reading messy paper is where AI helps; the pay / don't-pay rule must be testable and the same every time.
- **The invoice must agree with itself.** If the numbers don't add up, the agent knows it misread and doesn't guess.
- **Recommend, don't pay.** A person approves every payment; approving over a reject needs a written reason; a bank change needs a phone check first.
- **One question to the right person.** Every hold names an owner, the evidence, one specific question and a deadline that escalates.
- **Never reply to a possible fraudster.** For a bank change or look-alike vendor, the agent does not email the vendor.
- **n8n is the plumbing, not the brain.** A finance team can change who gets emailed without a developer; if n8n is down, nothing is lost, every message is in the Activity log.

Try it yourself (2 minutes)

1. Open the live app, press **Reset demo data** (top right), then go to **Upload & check**.
2. Pick **"Clean invoice, exact PO match"**: every stage passes and the agent says approve. Approve it under **Invoices**.
3. Pick **"Billed before all goods arrived"**: it holds and asks Stores. Open it and use **Try a reply (demo)** to answer as Stores; it re-checks.
4. Pick **"Real vendor, new bank account"**: it holds, and approve stays locked until you confirm a phone call.
5. Upload **your own invoice**: it is held as "vendor not onboarded" (correct). Press **Test setup** to create the vendor and PO as test data, or upload your own vendors, POs and goods receipts under **Records**.
6. See everything under **Reports**, and ask **Jeen** (top right), for example "What's waiting for me?".

Limits, and what comes next

- **Email only on my laptop.** n8n and Gmail run locally; the live link shows the messages instead of sending them. Next: host n8n alongside the app.
- **Test data, not a live ERP.** Vendors, POs and goods receipts are a realistic sample (or your own via CSV / Excel); the ERP posting is simulated. Next: connect Tally or SAP.
- **Government checks are offline.** The e-invoice IRN is recomputed and the QR decoded, but not checked against the GST portal. Next: live GSTN / IRP lookups.
- **Handwritten bills** are read only partly; very poor scans go to a person instead of being guessed.
- **Hosting** is a free trial with a small database that resets on restart: fine for a demo, not for production.

In one line
The agent does the checking and the chasing; a person makes the final call.